— *Solution notes* —

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

2 Artificial Intelligence (sbh11)

This questions concerns the *Partial-Order Planning Algorithm.*

Partial-order planning—ordering constraint and causal link.

(*a*) How does an *ordering constraint* differ from a *causal link*? [2 marks]

Answer: An ordering constraint between actions A_1 and A_2 specifies only that A_2 must happen at some point later than A_1 (1 mark). A causal link *also* indicates that action A_1 was added to achieve a specified precondition for A_2 (1 mark).

Partial-order planning—ordering constraint and causal link.

(*b*) Why is it necessary for the Partial-Order Planning Algorithm to consider both ordering constraints and causal links? [2 marks]

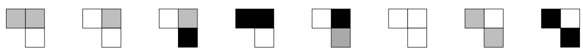
Answer: The causal link is needed to allow the algorithm to detect when an action A might interfere with an action A_2's precondition, which has been set up by an earlier action A_1 (1 mark), by occurring between A_1 and A_2 (1 mark).

A game involves filling a finite grid with coloured tiles. For a position (x, y) not on the first row or column, its *ancestors* are defined as positions $(x-1, y)$ and $(x, y-1)$, where the origin is at the bottom-left. For a given grid state, a tile can be placed at an empty position (x, y) if and only if:

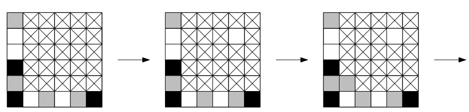
- its four adjacent squares are empty; or
- both ancestors have a tile, and the placement satisfies the Placement Rules; or
- only one ancestor has a tile, and the other ancestor can be filled later in a way consistent with the Placement Rules.

If a tile has been placed on every square, then the `Finish` state has been reached.

We will focus on a specific instance of the game with a 6-by-6 grid, where the colours are white, gray and black, and the Placement Rules are as follows:



In our instance the start state is fixed. Here are some possible moves from the start state, where a crossed square represents an empty grid position:



Solve the following problems using the Partial-Order Planning Algorithm.

Partial-order planning—start and finish states.

(*c*) Explain how the `Start` and `Finish` states can be represented for the instance given above. Only a description is required. [3 marks]

Answer: The x and y axes are labeled from 1 to 6. The `Start` state is a conjunction of literals, in this case indicating `clear`(x, y) whenever neither x nor y is 1 (1 mark). We also denote black, gray and white by B, G, W and include literals $\texttt{at}(1,1,B)$, $\texttt{at}(1,2,G)$, $\texttt{at}(2,1,W)$ and so on to reflect the initially placed tiles (1 mark). The `Finish` state can include literals with existentially quantified variables, so we just need to specify literals $\texttt{at}(2,2,z)$, $\texttt{at}(2,3,z)$ and so on covering all the grid spaces (x,y) where neither x nor y is 1 (1 mark).

Partial-order planning—actions

(*d*) Explain how actions can be represented modeling the placement of tiles, by giving examples based on the specific instance described above. [7 marks]

Answer: We use an action $\texttt{place}(x,y,z)$ (1 mark) with precondition $\texttt{clear}(x,y)$, $\texttt{clear}(x-1,y)$, $\texttt{clear}(x,y-1)$, $\texttt{clear}(x+1,y)$, $\texttt{clear}(x,y+1)$ (1 mark) and effect $\neg\texttt{clear}(x,y)$, $\texttt{at}(x,y,z)$ (1 mark) to model the placing of a tile when there are no squares around it occupied. The first rule shown above will need an action $\texttt{placeG1}(x,y)$ having precondition $\texttt{clear}(x,y)$, $\texttt{at}(x-1,y,G)$, $\texttt{at}(x,y-1,W)$ (1 mark), effect $\neg\texttt{clear}(x,y)$, $\texttt{at}(x,y,G)$ (1 mark) and further actions $\texttt{placeG2}(x,y)$ and so on for the other rules (1 mark). Finally we need to consider placement when only one ancestor is occupied, and this requires similar rules that can be enumerated by inspection (1 mark).

Partial-order planning—promotion and demotion

(*e*) Explain the concepts of *promotion* and *demotion*, which are of central importance in the Partial-Order Planning algorithm. To what extent might they be useful in the case of this particular algorithm and planning problem, given your formulation? Illustrate your answer with an example. [6 marks]

Answer: Each is the idea that if an action interferes with a causal link we can try to fix this by introducing a new ordering constraint (1 mark). We are likely to come across this in the current problem because all actions have $\texttt{clear}(x,y)$ in the precondition and $\neg\texttt{clear}(x,y)$ in the effect (1 mark). It is therefore likely that actions will interfere (1 mark). However in this particular example promotion or demotion will usually not help. Say we start by adding two actions to the initial plan: $\texttt{place}(4,5,B)$ and $\texttt{place}(4,4,B)$. These both rely on having empty squares around them, and will interfere. However promotion or demotion does not help, and we are forced to use different actions (2 marks).
